main.py

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Output

```
def bubble_sort_with_frames(arr):
    arr = arr.copy()
    frames = [arr.copy()]
    n = len(arr)

    for i in range(n):
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
        frames.append(arr.copy())

    return frames


if __name__ == "__main__":
    frames = bubble_sort_with_frames([5,1,4,2,8])

    assert frames[-1] == sorted([5,1,4,2,8])
    assert frames[0] == [5,1,4,2,8]
    assert len(frames) >= 2

    print("All test cases passed!")
```

```
All test cases passed!

=== Code Execution Successful ===
```